Technology, Knowledge and Learning (2025) 30:1847–1869
https://doi.org/10.1007/s10758-024-09780-z

ORIGINAL RESEARCH

Understanding Student Perceptions of Artificial Intelligence as a Teammate

Rebecca Marrone[1] · Andrew Zamecnik[1] · Srecko Joksimovic[1] · Jarrod Johnson[2] · Maarten De Laat[1]

Accepted: 26 August 2024 / Published online: 3 September 2024

Abstract

This article examines students' opinions regarding the use of artificial intelligence (AI) as a teammate in solving complex problems. The overarching goal of the study is to explore the effectiveness of AI as a collaborative partner in educational settings. In the study, 15 groups of grade 9 students (59 students total) were assigned a challenging problem related to space exploration and were given access to an AI teammate. Following the task, the students participated in focus group discussions to gain insight into their perspectives on collaborating with AI. These discussions were analysed using thematic analysis to identify key themes. Epistemic Network Analysis was then used to quantify and visualise this data. The results suggest that students perceive AI with regard to two main themes: Trust in AI and the Capability of AI. The study's outcomes shed light on how students perceive AI and provide practical recommendations for educators to effectively incorporate AI into classrooms. Specifically, the recommendations include strategies for building student trust in AI systems through Explainable AI processes. This, in turn, encourages collaboration between humans and AI and promotes the development of AI literacy among students. The findings of this study are a valuable addition to the ongoing discussion on AI in education and offer actionable insights for educators to navigate the integration of AI technologies in support of student learning and growth. The scientific contribution of this study lies in its empirical investigation of student-AI interaction, providing evidence-based insights for enhancing educational practices.

Keywords Human-AI teaming · Complex problem solving · Artificial intelligence in education · Student perceptions of AI

✉ Rebecca Marrone
Rebecca.Marrone@unisa.edu.au

[1] The Centre for Change and Complexity in Learning, University of South Australia, Adelaide, Australia

[2] Pulteney Grammar School, Adelaide, Australia

1 Introduction

The integration of Artificial Intelligence (AI) in education systems has become increasingly prominent, as it transforms the educational landscape by personalizing learning experiences, automating administrative tasks, and enhancing educational outcomes (Zawacki-Richter et al., 2019). Despite the prevalence of AI in students' daily lives through platforms such as Snapchat, Siri, and Alexa, to name a few, they often fail to recognize when they are engaging with AI-enhanced systems, highlighting the lack of understanding of AI's capabilities (Druga et al., 2017, 2018). This limited knowledge of AI may result in misconceptions, critical or limited effective use, and disappointment if expectations are not met (Crompton et al., 2022).

While AI is gaining traction in many industries, supporting workers in automating manufacturing tasks, assisting in learning and teaching, and enhancing human creativity (Marrone et al., 2022), its integration into human teams is still an emerging area that requires attention and further research. As AI develops, its adoption in team environments is only set to increase in the workplace, highlighting the need for students to develop human-AI teaming skills for their future success (Zhang et al., 2021).

The increasing complexity of modern life requires multidisciplinary and intercultural cooperation to tackle complex problems that demand coordination within a multidisciplinary team (Ekbia & Nardi, 2017; De Laat, Joksimovic, Ifenthaler, 2020). There has been extensive research in the fields of team science (Cooke & Hilton, 2015) and collaborative learning (Jeong et al., 2019) which need to be integrated into frameworks of human-AI teaming (Seeber et al., 2018), and in particular student-AI teaming. Studies on team science emphasize the importance of communication, trust, and shared goals in successful collaboration, which are equally relevant in human-AI teaming (Salas et al., 2008). When engaged in complex problem solving, the specific roles of humans and machine will require ongoing negotiation. AI will (over longer periods of time) learn to better understand and meet the needs of human teammates. Correspondingly, humans will adapt what they do as they learn to better use what the AI team has to offer. This notion of human-AI teaming as a concept that emerges through active negotiation and other forms of mutual adaptation will raise the need for new skills and literacies regarding cognition and collaboration between the two systems (Siemens et al., 2022). For example, students might need to learn how to interpret AI-generated data and make decisions based on AI recommendations, which requires critical thinking and technical literacy (Walter, 2024).

There are a number of emerging studies on student perception of AI. One study suggests that students develop a more positive attitude to learn AI and enhance teachers knowledge of AI when an AI based curriculum was co-created with university professors (Chiu et al., 2021). Another study identified factors that influence primary school students perceptions of AI. The findings revealed that AI literacy alone did not determine the readiness of individuals towards AI. Instead, the impact of AI literacy was indirectly influenced by the level of confidence and the perceived significance of AI as perceived by the students (Dai et al., 2020). Despite this research, there remains a gap in how students perceive AI as a teammate. Addressing this gap is crucial for designing effective human AI teams in the classroom (Holmes et al., 2019).

The study aims to contribute to the growing body of knowledge regarding AI in education and provide insights on how educators can integrate AI to improve student learning outcomes. Focus groups were conducted to gather data on what students think about working with AI and provide practical recommendations for educators to implement AI in their classrooms to support human-AI teaming and the development of critical AI literacy. These recommendations include training programs for both students and teachers to enhance AI literacy, integrating AI projects into the curriculum, and fostering an environment that encourages experimentation and collaboration with AI technologies (Touretzky et al., 2019)(Touretzky et al., 2019). This paper, therefore, aims to answer the following research question: What are the perceptions of students after working with artificial intelligence to solve a complex problem together?

2 Background

2.1 AI in Education

In the context of education, AI typically refers to a suite of technologies that enable machines to perform tasks associated with human intelligence, such as recognizing speech, supporting decision-making, and solving problems. Student and teachers interact with generative AI to consume and create content for assignments (BaiDoo-Anu & Owusu Ansah, 2023). AI technologies in education are now applied in a variety of educational settings to enhance teaching and learning, providing benefits such as personalised instruction, for instance, where AI is used to adapt the pace, content, and assessment to learners' individual needs (Chiu, 2021; Dilmurod & Fazliddin, 2021). AI has also been used to support teachers by automating routine administrative tasks or grading, allowing them to focus on more complex teaching strategies and engage more with students (Ahmad et al., 2022; Seo et al., 2021). Finally, among other applications, AI can help scale education to meet the needs of a growing student population, particularly in areas where there are shortages of qualified teachers (Alam, 2021; Chounta et al., 2021). This can help ensure that more students have access to high-quality education, regardless of their geographic location (Alam, 2021).

While AI has shown great potential in enhancing education, recent reports suggest that there are still significant challenges to be addressed. Ferguson et al., (2016) note that there is a lack of consistent updates and systems-level implementation of intelligent educational tools in the classroom. Additionally, Freeman et al., (2017) point out that individual schools may not be integrating AI technologies into their curricula or providing sufficient opportunities for teachers to upskill in this area. Chounta et al. (2021) also argue that the available AI-enhanced technologies do not adequately address the needs of K-12 teachers and may not be widely adopted as a result. These issues may stem in part from the fact that many AI technologies currently used in the classroom focus on individual student or teacher support, potentially isolating students and hindering collaboration and teamwork. However, an emerging area of research and development is focused on human-AI teaming, in which AI technologies are designed to work in tandem with human educators and students. By leveraging the strengths of both humans and machines, these systems have the potential to revolutionize education and improve student outcomes (Luckin & Holmes, 2016).

2.2 Human-AI Teaming

Humans and AI agents collaborating towards achieving a common goal is frequently referred to as human-AI teaming (Ezer et al., 2019). This collaboration can take various configurations, from human-AI dyads, where a single human works with an AI agent, to teams where multiple humans work with one or more AI agents (Ezer et al., 2019; Zhang et al., 2021). Although AI research has not yet produced technology capable of engaging in the cognitive (e.g., a shared mental model), and social (e.g., trust and relational) processes needed for critical thinking and problem solving that are on par with human abilities, considerable progress is being made recently toward those goals (Amershi et al., 2019; Seeber et al., 2018).

In the context of education, intelligent tutoring systems exemplify human-AI teaming by personalizing the learning experience and adapting to the student's needs and progress (Chen et al., 2022; Mousavinasab et al., 2021). These systems use AI algorithms to provide feedback and guidance, helping learners overcome challenges and achieve their goals (AlShaikh & Hewahi, 2021). Specifically, despite Bayesian Knowledge Tracing being a dominant approach in student modeling, recent studies showed comparable or improved performances using Deep Learning approaches (Minn, 2022). This represents a form of human-AI teaming where the AI supports the human learner through adaptive interactions. Social robots can also provide personalized instruction and feedback in an arguably more engaging and interactive way (Papadopoulos et al., 2020). For example, social robots can be used in language learning to provide conversational practice and feedback, or in STEM education to provide hands-on demonstrations and explanations (Ahmad et al., 2020; Vogt et al., 2019). These interactions illustrate how human-AI teaming can enhance educational experiences by leveraging the strengths of both human teachers and AI. Finally, AI-powered personal assistants, such as Alexa and Siri, can also assist with learning tasks, that include searching for information and organizing notes (Hsu et al., 2021). These assistants exemplify another facet of human-AI teaming, where AI aids in managing and supporting educational tasks, thereby enhancing the overall learning process.

With the recent developments in Generative AI, a plethora of AI-enhanced tools is emerging almost daily, primarily aimed at enhancing individual's productivity. Such advances are continuously challenging our previous conceptions of learning, knowledge and education in general. As (Lodge et al., 2023) suggest, "there is not a single metaphor for the relationship between human and AI in learning, but many" (Page 1).

In this paper, however, we particularly focus on AI agents that are capable of working in teams. As such, this work is closer to the Fiore's conceptualisation of Artificial Social Intelligence (Fiore, 2021; Williams et al., 2022), or the notion of using AI in complex problem solving, as outlined by Joksimovic and colleagues (2023). In this context, AI has been commonly deployed to support team roles and task roles. For instance, AI can assist with information gathering and processing, allowing team members to focus on more complex and creative tasks (Joksimovic et al., 2023). In addition, AI can support team coordination and communication by providing real-time updates and facilitating collaboration and cooperation (Musick et al., 2021).

There is a growing body of research dedicated to exploring how human and artificial teaming can support specific learning domains, including complex problem solving (Bayrak et al., 2021; Joksimovic et al., 2023; Schaefer et al., 2021). Virtual assistants, such as ChatGPT, can be designed to assist students in developing their communication skills by providing feedback on their writing and helping them refine their arguments (BaïDoo-Anu & Owusu Ansah, 2023). De Laat and colleagues (2020) have shown that human-AI collaboration in complex problem-solving has been investigated across a wide range of AI application domains. However, AI has augmented the four dimensions of complex problems—cognitive, metacognitive, social, and affective—to different extents. While most research has focused on the cognitive domain, progress is being made in social and affective dimensions as well (e.g., Joksimovic et al., 2023).

However, human and artificial teaming in education presents several challenges that must be addressed to ensure effective and ethical use of AI (Textor et al., 2022). Among these challenges is the issue of trust, as some individuals may be sceptical of using AI systems in the classroom due to concerns about their reliability and potential biases (De Visser et al., 2020). To overcome these challenges, researchers have proposed a range of design principles for human and artificial teaming in education (McCall et al., 2021; Renz & Vladova, 2021). These principles emphasize the importance of transparency and accountability, ensuring that AI systems are designed in a way that makes it clear how decisions are being made and how the systems can be held accountable for their actions (Lyons et al., 2019; McNeese et al., 2021; Textor et al., 2022). Flexibility and adaptability are also key principles, allowing AI systems to be adjusted to meet the changing needs of students and teachers.

2.3 Perceptions of AI as a Teammate

While there is evidence that AI can support business outcomes (see Wilson & Daugherty, 2018), Zhang et al. (2021) demonstrate that little research investigates how people *perceive* AI as a teammate. Zhang et al. (2021) surveyed 213 participants and conducted follow-up interviews with 20 participants. The results highlight that people have mixed feelings toward AI teammates but hold a positive attitude toward future collaboration with AI teammates. Moreover, Zhang et al. (2021) explain that the expectation of an AI teammate to perform to the same standard as a human teammate is, in most fields, impossible, or perhaps not needed at the first place. However, people have incredibly high and unrealistic expectations of what AI is capable of. Holstein and Aleven, (2022) worked with teachers to co design an AI application that helps teachers track student learning through the use of smart glasses. Similarly, (Wang et al., 2022) showed how AI can be seen as a coach for second language learning in primary school students.

In their study, Holstein and Aleven (2022) note that those students learnt more when teachers and AI tutors worked together during class, however, they attribute the results to the participatory design nature of the project. Whilst the inclusion of teacher voices is crucial to AI integration equally important is understanding student perceptions.

In their meta-review Fast and Horvitz (2017) reported that the most common concerns reported from the general public about AI are: (1) the fear losing control over the machines and (2) the lack of ethics concerning AI development to the point that human life, and humanity altogether, would be threatened. Van Brummelen et al., (2021) further investigated how middle and high school students' perceptions of Alexa changed through programming their own conversational agents in week-long AI education workshops. The authors found that students felt Alexa was more intelligent and felt closer to Alexa after the workshops. They also found strong correlations between students' perceptions of Alexa's friendliness and trustworthiness, and safeness and trustworthiness after interacting with AI. Whilst an interesting finding, (Karampelas, 2020) showed that students report little to no prior interaction with AI. Finally, Marrone et al., (2022) demonstrate that mere exposure to AI is not enough to alter student perceptions, and suggest effort needs to be made to address myths and misconceptions.

This paper focuses on the perceptions of students after working as a team with AI in a problem-based learning context. Given the lack of current studies that explore the perceptions of students after working with AI, there is a need to understand how AI impacts their learning and what educational stakeholders can expect when AI is involved in the classroom, such as its benefits and drawbacks. As a result, the study aims to answer the following research question: What are the perceptions of students after working with artificial intelligence to solve a complex problem together?

3 Method

3.1 Program Structure

This study took place at an Australian private co-educational school where K-12 students are studying towards their certificate of education in a blended learning environment. These environments allow for streamlined instructions created by the teacher to help with learning at scale, supporting access to ‘out of hours’ materials through technology and personalised learning to cater for the student’s individual needs (Staker & Horn, 2014). As part of the school’s curriculum, the students were required to participate in a complex problem-based learning context and work as a team to build 21st-century skills, namely, collaboration, problem-solving and communication skills over a semester, approximately 12 weeks.

3.2 Study Design and Participants

There were 59 students in grade 9, approximate age of 15–16 years old who were divided into 15 teams in shared leadership roles containing two to five members per team. Teams were pre selected by the school based on teacher professional judgement. The teacher opted to have ‘higher ability’ based teams and ‘lower ability’ based teams so appropriate scaffolds could be implemented during the period. Lower ability based teams were approximately C-grade students, and higher Ability based teams were A or B grade students. There were no students who were considered as ‘failing’ students. Students have a normal distribution of additional learning needs and requirements. Teams participated in a semester-long blended learning environment working on two major challenges in their study. In the first challenge, teams had to work together to solve complex problems related to “the future of work”, such as the challenges of living on the planet Mars and colonising planets. In the second and final challenge, they worked with AI in the “AI Playground” to build a Mars Rover and determine how to correctly modify and demonstrate the Mars Rover’s performance on a simulated Martian surface, getting from point A to point B successfully. The assessments have been structured into specific deliverables: (1) working together as a team; (2) presenting their work; and (3) completing their projects that match specific learning outcomes required from the certificate of education.

The focus of this paper concerns the second part of their challenge offered by the AI Playground, where students worked collaboratively with an AI agent—Viana—that provided them with real-time feedback when building a Mars Rover. Viana is a Visual AI agent that uses computer vision to view the progress of the Mars Rover through Image Recognition and is trained through a neural network (deep learning) algorithm. Viana, as a team member, had two roles: (1) to provide feedback and recommendation in building the next component for the Mars Rover; and (2) to provide a progress report for the Mars Rover. Viana has been co-designed and created by an industry partner specializing in AI products. For example, as students worked on assembling the Rover's chasis, Viana identified potential misalignments or errors in real-time. If a student placed a component incorrectly, Viana would alert them with a visual prompt and suggest corrective actions. Viana's feedback was not limited to error correction; it also included optimization suggestions. For instance, when students were deciding between different types of materials for the Rover, Viana analyzed the options and recommended the most efficient option based on the mission requirements and previous data. Additionally, Viana provided periodic progress reports, summarizing the Rover's assembly status, highlighting completed milestones, and outlining the next steps. This interaction not only guided the students through the technical aspects of building the Rover but also helped them understand the decision-making process and the rationale behind each recommendation, fostering a deeper engagement and learning experience.

The student teams are required to build the Mars Rover alongside Viana by building the product and regularly checking the feedback and progress provided by Viana. Students do not have access to a hard copy of the manual to build the Mars Rover for this exercise. They entirely rely upon the accuracy of the Visual AI agent. Students were instructed to think of Viana as a teammate. Before conducting the research, the university's internal review board granted the appropriate procedures requiring ethics clearance to collect and analyse the data, which extended to surveys and focus groups.

This study focuses on how the team members worked alongside Viana in a complex problem-based learning setting. We collected and conducted ($n=12$) focus groups with a total ($n=50$) team members to gather their perceptions about AI and how educational stakeholders can apply this information to inform their decision-making processes when implementing AI into the curriculum. These transcribed data contained a total count of 8117 words in the text. The minimum word response to the focus group questions ranged from 1 to a maximum of 264, with a mean of 45 words. Finally, Table 1 shows the main six focus group questions to gather information about AI perceptions.

3.3 Analytical Approach to Examine Perceptions

To answer our research question, we conducted a mixed method approach using Epistemic Network Analysis (ENA) to extract patterns (Shaffer et al., 2016) and an inductive reflexive thematic analysis to extract themes (Braun & Clarke, 2019) about the perception of AI in a K-12 setting. ENA is commonly used in educational research as a method for quantified links among coded data in a dynamic network model (Shaffer et al., 2009). ENA is also used to identify patterns of association, structures and strengths in a complex network

Table 1 Focus group questions

Question ID	Question names
Q1	What is AI?
Q2	How do you feel about AI?
Q3	Would you want to work with AI?
Q4	Do you think AI can match human skills or replace them?
Q5	What skills are important for the future of work?
Q6	How was your experience working with AI?

over time (Shaffer et al., 2009). The X and Y axis in an ENA Plot accounts for variance in the data. The axis with a higher percentage amount indicates that the dimension is highly informative about the structure of the data, resulting in more effective visualisations of ENA.

For the thematic analysis, we followed the six phases outlined by Braun and Clarke (2006), namely: (1) familiarisation with the data; (2) generating initial codes; (3) searching for themes; (4) reviewing themes; (5) defining and naming themes and; (6) producing the report. The reflexive thematic analysis is an approach that is flexible and not bound to specific frameworks and allows the experience of the researcher as the primary means to discern knowledge from data (Braun & Clarke, 2019). We conducted focus group interviews with grade 9 students to collect transcript data regarding their perceptions of AI. As such, six open-ended questions were given to the focus groups. These questions in the focus group are based on Table 1. Each response made by the focus groups was coded, linking to a common theme. Two reviewers calculated the Inter-rater agreement (IRR) and Kappa to reach an appropriate level of consistency and reliability. The same reviewers were also involved in conducting the focus groups. Codes were adjusted and iterated to ensure robustness through discussion between two reviewers in phase three of the thematic analysis process. For example, ensuring that the interpretation of the transcript data between the two reviewers are meaningful and accurate.

4 Results

4.1 Coding of Transcript

In Table 2 below, an excerpt is provided to show how the transcript data was segmented. The Stanza column is used to quantify relationships between objects and is used to measure code co-occurrence. The Unit column is used for objects to understand interactions between codes. The Code columns contain a set of concepts where we wish to understand the relationships between them.

The responses of all focus groups for each question were collected with their respective codes and split into separate CSV files. These files were formatted to be consistent with a co-occurrence table as input data for epistemic network analysis.

4.2 Interrater Agreement

Two reviewers coded the 34 codes during phase 2 of the thematic analysis procedure (Braun & Clarke, 2006) with multiple iterations, which resulted in the IRR for questions Q1 to Q6 with an average of 92% accuracy and an average 0.80 using Cohen's Kappa (Cohen, 1960). The reviewers coded all codes for each question. Conflicts were discussed and resolved when generating codes to reach a Kappa of 0.70. The associated reliability measures for each question and their codes are found in Table 3.

4.3 K-12 Perceptions in AI

The analysis conducted by the two reviewers generated 34 codes in total for the six questions. The results of the coding scheme table for each question conducted by the two

Table 2 Excerpt of a coded spreadsheet file containing responses from focus groups

Metadata columns				Code columns					
Stanza column		Unit column	Raw data column						
Line	Activity	Group	Text	Learning Impact	Openness	Apprehensive	Teaming	Adaptiveness	Brand Famili-arity
1	Would you want to work with AI?	Focus Group 12	*"Humans because you can interact better with them because it's more, you can't relate to a robot in any way"*		1	1			
2	Would you want to work with AI?	Focus Group 12	*"Yeah, I think, yeah, I would get a different outlook. I feel like I don't know. I don't think artificial intelligence could have like ethics. Mm. So prob-ably not."*		1	1	1		
3	Would you want to work with AI?	Focus Group 13	*"I guess I would, because if I do get stuff that I have something that'll help me and support me through whatever I'm doing to it. I mean, I probably only use if Apple made it. Oh, of course"*	1				1	1

Table 3 Inter-rater agreement and Cohen's Kappa

Question ID	IRR agreements (%)	Cohen's kappa
Q1	93	0.80
Q2	91	0.80
Q3	94	0.85
Q4	90	0.78
Q5	94	0.85
Q6	87	0.74

reviewers is shown in Table 4. It is important to note that some code-naming conventions and descriptions overlap.

Six ENA projections were created for each question based on the relative codes outlined in the coding schema for Table 4. Figure 1. shows the coded co-occurrence relationships of the results for each question, where thicker lines indicate connections that occur more frequently, and thinner lines indicate fewer connections. Nodes positioned closer mean less variance between them than nodes farther apart (Shaffer et al., 2016). Figure 1A (i.e., Q1: What is AI?) suggests that students can provide a relative description of AI based on the strong connections between *Awareness* and *Portrayal* nodes. These relationships also extend to their knowledge of AI *Impact*. It seems that students are capable of inferring the potential of AI and how it could disrupt and change their learning. The *Future* and *Anxiety* codes often linked with *Portrayal and Sentiment*. There is a lack of connections between *Anxiety* and *Impact, which* might indicate that students have unrealistic expectations of the capabilities of AI.

Figure 1B (i.e., Q2: How do you feel about AI?) shows the relationship between students' feelings towards AI. The results reveal that students have an overall positive feeling about AI that depends on the pedagogical context and situation based on the strong connections between *Education*, *Sanguine* and *Disposition.* Students also shared how they contextualise the impacts of AI in the educational landscape depending on the context, as indicated by the connections with *Relatability*, *Disposition* and *Education*. Overall students have a positive sentiment about how AI can impact their learning.

Figure 1C (i.e., Q3: Would you want to work with AI?) shows that student's express willingness to work with AI and believe it will positively impact their learning based on the strong connections between *Assisted Learning* and *Adoption*. The codes have moderate connections with *Trust Task*, where students made examples of trusting AI in certain situations. In retrospect, the code *Scepticism* shares relative connections with *Teaming and Adoption*, indicating certain situations where AI may not be trustworthy. The code *Product Affinity* has small connections between *Assisted Learning* and *Teaming*, suggesting that students may be particular in what kind of AI products they choose to work with.

Figure 1D (i.e., Q4: Do you think AI can match human skills or replace them?) shows the results of student perceptions about AI matching or replacing human skills. Most students indicated that AI would match or overtake human-related skills in the future, as shown by the strong links with *Skill Substitution* code. The links between *Skill Substitution, Human Skills* and *Affective States* indicate that AI can match human emotions in the future. Although there are stronger connections between *Limitations*, *Affective States* and *Human*, most students believe AI may not achieve this outcome. Moreover, *Misuse* and *Uncertain* connections with *Affective States, Limitations and Skill Substitution* will be mediated by the students' trust in AI and their uncertainty to a lesser extent. In other words, even if AI

Table 4 Coding scheme and description

Question ID	Codes	Code descriptions
Q1	1. Awareness	Knowing what artificial intelligence is
	2. Sentiment	Related to the sentiment of AI
	3. Impact	What AI can do in education and the world
	4. Anxiety	unsure about a future with AI
	5. Portrayal	How is AI viewed (i.e., machine, robot, program)
	6. Future	What future and role AI has in the world
Q2	7. Sanguine	Optimistic sentiment of AI
	8. Relatable	How students relate to AI
	9. Education	How AI can benefit pedagogy
	10. Disposition	Context of AI and trustworthiness
Q3	11. Assisted Learning	How AI impacted the students learning
	12. Adoption	Willingness to work with AI
	13. Skepticism	Negative feelings towards AI
	14. Teaming	human-autonomous teaming
	15. Trusted Task	Examples of trusting AI in certain tasks
	16. Product Affinity	Only working with specific AI products
Q4	17. Skill Substitution	Compatible in matching or replacing human skills
	18. Human	Example of human only skills
	19. Technology	Example of technology skills
	20. Limitations	AI cannot match or replace humans skills
	21. Affective States	Examples of emotions, feelings and an understanding
	22. Uncertainty	Unsure if AI can replace or match humans
	23. Misuse	AI becoming untrustworthy

Table 4 (continued)

Question ID	Codes	Code descriptions
Q5	24. Social Skills	Soft skills focusing on emotion, personality, communication,
	25. Hard skills	Hard skills focusing on programming, coding, AI
	26. Problem-solving	Soft skills such as problem-solving
	27. Creativity	Soft skills such as creativity
	28. Storytelling	Soft skills such as storytelling
	29. Traditional skills	Skills that are traditional (i.e., writing)
Q6	30. Fun	How fun it was to work with AI
	31. Performance Issues	The AI product suffering from glitches or breaking down
	32. Maintenance Issues	Having to problem solve and fix the AI product
	33. Educating	It helped with students learning and motivated them
	34. Dissatisfaction	Reported frustrations, not wanting to work with AI

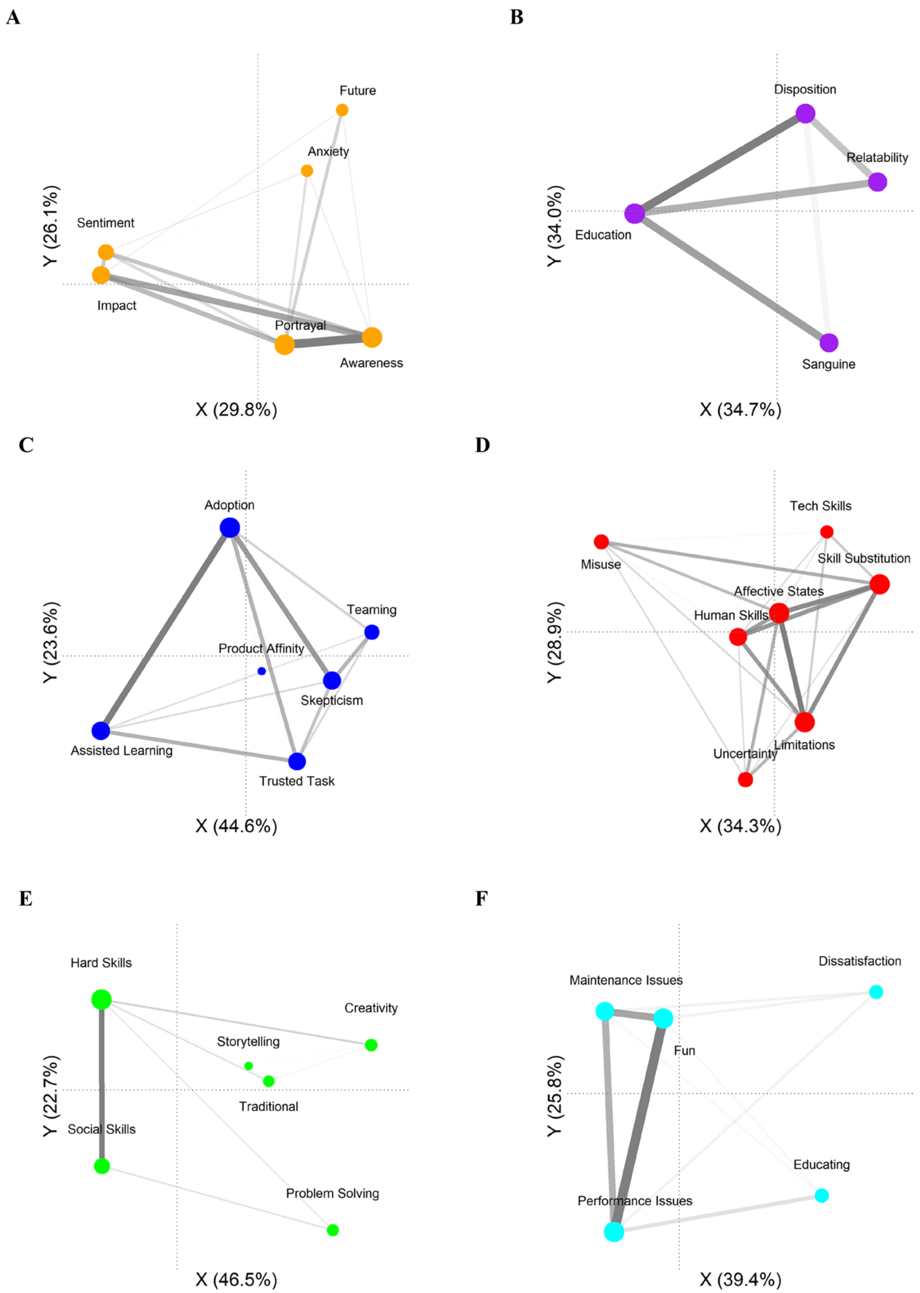

Fig. 1 The plot contains the projection of the six questions (Table 4) and their respective codes rotated by their respective means

can achieve replacing or matching skills, there are indicators where students are still wary about AI, such as the working context and activities the AI is set to do.

Figure 1E (i.e., Q5: Skills that are important for future of work?) shows the results of what students perceive as important skills for the future. The *Hard Skills* and *Social Skills* codes have strong links, indicating that students believe a combination of hard skills, (i.e., programming, coding, working with AI and other technical applications) and soft skills (i.e., social interactions, personalities and emotions) will be important. Moreover, students gave examples of soft skills such as *Problem solving*, *Creativity* and *Traditional* that will be important for the future of work. These results suggest that students recognise the importance of obtaining skillsets where AI cannot compete with them and will remain crucial in the workplace.

Figure 1F (i.e. Q6: How was your experience working with AI?) shows the students' perceptions and relationships when working with AI. As such, many students experienced difficulties working with AI (i.e., the strong connections between *Performance and Maintenance Issues*). Moreover, these difficulties had strong links with *Fun* and some links with the *Educating* codes, suggesting that even though students had problems working with the AI, the experience was still enjoyable and motivating for their learning. There were a small number of instances where some students were frustrated or rejected the idea of working with AI (i.e., *Dissatisfaction*), often linked with experiences such as *Performance and Maintenance Issues*.

As a result of the ENA and thematic approach, two themes emerged: Trust in AI and Capability of AI, which dominate the student perceptions of AI. The themes generated were carried out by turning the identified codes from transcripts into reflexive themes. Codes that were not generalisable were discarded. The six ENA projections aided in the development of our four themes, which contributed to the narrative of the data. In Fig. 2, the colour-coded solid lines represent how each ENA projection was fostered in the four themes' development. The final thematic framework mapping is shown in Fig. 2.

4.4 Theme 1: Trust in AI

The results suggest that students generally accept AI. They perceive AI as being *"cool"* and *"useful"* (Focus Group 15) but scary simultaneously, as they fear they will one day be replaced or lose their jobs: "*Employees are becoming a lot less important now"*—(Focus Group 1). Another quote: *"I don't think any human wants the world to be run by AI. It is already overtaking humans and things like working in factories and stuff*" – (Focus Group 14). The student's sentiments when working with AI depended entirely on hypothetical contexts. For instance, opposing AI as a leader (vertical leadership). Students wanted to be challenged to learn and not be dictated by AI: "*Because I feel like for me, I'm still going to try to like not listen to AI. I'd also try it out of my own information to it"*—(Focus Group 11). Many students do not trust AI being a co-member of a team (horizontal leadership), as they know it is not human: "*It's just whether the team can listen to the AI as if it were an actual person*"—(Focus Group 11). AI is accepted more as a function to help automate mundane experiences, used as a tool at most. Students do not recognise AI as a living or human being because it takes away a living thing's social and emotional aspects. One student described it as such: "*That's still not genuine emotions. It's not like they're actually thinking and, they can just be programmed to be your friend.*"—(Focus Group 8) Some students reported that they would only work with AI if made by certain manufacturers: *"I probably only use it if Apple made it"*—(Focus Group 13). Overall, students would rather

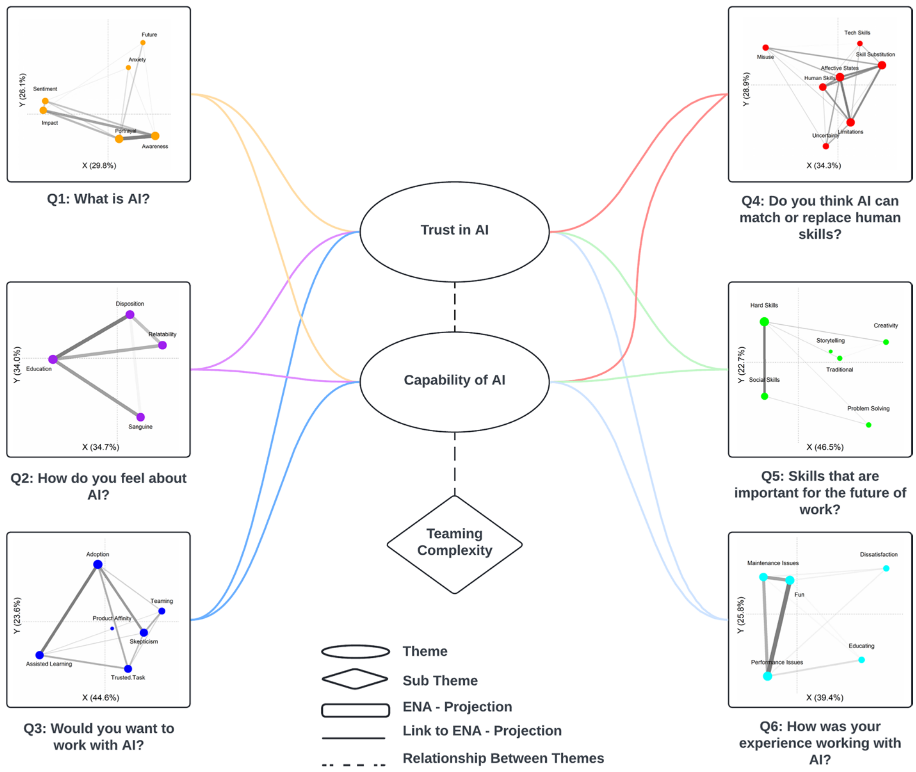

Fig. 2 Finalised thematic map demonstrating our two themes based on the six ENA projections

work with humans than robots in most situations as described by a student: "*Because you can interact better with them, (humans) you can't relate to a robot in any way"*—(Focus Group 8).

4.5 Theme 2: Capability of AI

Most students provided a technological description of AI with the following: "*artificial intelligence is pretty much just robots kind of learning, and they kind of adapt based on what you do to either help a person do something or to make it more difficult for a person to do something*"—(Focus Group 7). AI also will eventually be smarter than humans: "*AI will have a bigger scheme of knowledge than humans do*"—(Focus Group 8). This perception that AI is a "*robot*" or "*super intelligence*" may be attributed to the influence of marketing and social media. Other influences where students had drawn upon ideas about AI are from documentaries: "*Oh yeah, we watched a documentary on it*", movies: "*My mind just goes to Jarvis from Avengers.*"—(Focus Group 12) and AI companies such as Google, Microsoft and Amazon: "*So things like Alexa might just do my homework for me*"—(Focus Group 15). Overall, students reported an understanding of what AI might be in the future but not what AI is currently capable of in the context of their learning.

Students have a broad view of what AI can bring into the educational landscape, from simple to complex tasks. For the former, using AI as a calculator to solve math equations:

"*Tell me your math equations. Solve this calculus for me?*"—(Focus Group 15), and for the latter, using AI to act as an additional teacher that personalised the learning of the student and caters for their needs: *"A personal one (AI) for each person in the classroom that can help them kind of go with their speed. We've got some people who are excelling and some people who are still learning. It very difficult for the teacher to be both kind of compensating to the people who are just beginning and the people who are all the way there already"*—(Focus Group 7). Another impact that AI has is that it prompts students to think about the future of work and what skills they will need. As such, students do not want to have the skills they learned to be made redundant by AI: "*We are all worried that artificial intelligence is most likely going to take over our jobs and humanity*"—(Focus Group 12). There are a few examples of emphasising the importance of "Non-Googleable skills". Namely, problem-solving, communication and creativity skills: "*The soft skills is more important than, let's just say, knowing how to use technology"* – (Focus Group 8). Students want to obtain skills that AI cannot match and replace.

4.6 Sub Theme 2: Teaming Complexity

Some frustrations were reported when students worked with Viana. Given that AI tends to provide inconsistent results and is prone to errors or mistakes: "*Sometimes you accidentally scan the thing that you're trying to copy and then it would say 100% complete*"—(Focus Group 11). Students often would seek solutions from others: "*I'd prefer someone else there because they could tell you what you've done wrong"*—(Focus Group 12) or try to work it out themselves when solving problems with AI. They also understand that AI has weaknesses and sometimes need to take the initiative and avoid relying upon AI to do the work for them. For instance, students would resort to reading a more comprehensive manual for solutions. Nevertheless, they found it engaging and fun to work with: "*I enjoyed building it, it was fun. A bit of a change working on our computers"*—(Focus Group 6). *It was a good experience getting to do that because it was something that is kind of a new thing that we haven't really done before*"—(Focus Group 7). There was a notable example made that if students could work with AI in a way that could remove "sheet work" or "computer work", it would help optimise their learning: "*I thought that was a good way to transfer information instead of just like some task? sheets actually critically thinking. I feel like that it really leaves a good mark in understanding in our learning*"—(Focus Group 4). Overall, the focus groups often highlighted the impacts of AI on learning and the future of work, how AI impacts their sentiment and trust based on situational scenarios and the challenges of making AI work in educational environments.

5 Discussion

5.1 Themes

Thematic analysis of the 15 focus groups indicate that students view AI in regards of 2 key themes: Trust in AI, and capability of AI. Whilst there were overlaps between the 2 themes students clearly reported a variety of perceptions towards AI.

Students reported that AI could help them in their current classroom through differentiation practices and assistance with homework, which is consistent with some of the previous work, such as Demir and Güraksin, (2022). However, students also reported that there was some fear and uncertainty in how AI will impact their ability to enter the workforce. This finding is also not surprising, given that Keles and Aydin, (2021), among others, showed that negative perceptions of AI tend to be more significant than positive. Notably, students emphasized the importance of soft skills, which are essential for distinguishing humans from AI in the workforce. Such perception can likely be explained with the dominant narrative in the current research and mainstream media (Kaput, 2022). This highlights the need for educators to prioritize AI literacy as a cornerstone for future teaching and learning, as well as to incorporate soft skill training into the curriculum. By doing so, educators can not only help students develop the skills that AI cannot replace, but also demonstrate how AI can be used to enhance and develop these critical soft skills.

5.2 Trust in AI

Students tended to trust AI applications from certain companies they have already engaged in and therefore have experience using their products (e.g., *I probably only use it if Apple made it"*—(Focus Group 13). However, as AI applications increased in complexity, the trust in the system diminishes. This probably pertains to the idea of Explainable AI, which involves creating a model that is transparent. That is, the idea of Explainable AI is based on the principle of creating machine learning models that can be understood by humans (Gunning & Aha, 2019; Holzinger et al., 2022). In other words, the model's decision-making process is transparent and can be easily explained. This is important for ensuring that the model's decisions are fair, unbiased, and can be trusted by users (Pearl, 2019). Although transparency has been considered to be crucial for establishing trust in a system, recent studies show that it might not be critical. Specifically, with the emergence of Generative AI and ChatGPT in particular, wide population readily engaged with AI, without necessarily understanding how this system might work. Perhaps, the notion of low stakes and high stakes situations can explain this phenomenon, where for example, students felt comfortable with AI taking over factory jobs but would not feel comfortable with AI '*flying a plane'* Focus Group 3. This supports previous work that shows that people are more likely to trust AIs with high-performance than with low-performance (Yin et al., 2019; Zhang et al., 2021).

5.3 Capability of AI

The present study found that students possess unrealistic expectations of AI and exhibit low levels of AI literacy, as evidenced by their primary reference point for defining AI being pop culture figures such as "*Jarvis from the Avengers*"—(Focus Group 10). Despite receiving AI training, students still maintained strong misconceptions, indicating a need for further educational interventions to improve their understanding of AI's capabilities. Previous research has shown that humans often hold unrealistic expectations of AI across various domains, despite AI's advancements in machine learning algorithms that have enabled it to outperform humans in some tasks (Zhang et al., 2021). It is important to note, however, that AI has limitations and cannot perform identically to humans in all respects. Zhang et al., (2021) suggest that human expectations of AI may be influenced by experiences collaborating with humans in games, leading them to replace human partners with AI and expect it to possess all of the positive qualities of humans while also surpassing them. In light of this, human-AI teaming designers may need to shift their focus from creating an ideal human substitute to developing an ideal AI teammate. Future research should concentrate on designing tasks and lesson plans that involve co-creation with students and clearly explain AI's capabilities to improve their AI literacy and trust in AI, ultimately leading to more successful outcomes (Crompton et al., 2022).

5.4 Complexities with AI

Students reported that the main complexity issue they experienced with AI was frustration when it did not perform as well as a human, which has been previously reported in research (Markauskaite et al., 2022). The students also acknowledged that AI is riddled with errors and cannot perform perfectly E.g., "*Sometimes you accidentally scan the thing that you're trying to copy and then it (Viana) would say (the rover) was 100% complete*" – (Focus Group 11). It is important to not overemphasize the promises of AI. In order for AI to be beneficial for learning, it needs to go beyond the delivery of simple tasks that can be easily repeated without it, as it would be more trouble than it is worth. There should be clear benefits for using AI, otherwise it is not necessary. Although students recognize that AI has the potential to impact education and learning in great ways, it is essential to provide training on AI before its introduction, as there are wild perceptions of what it can and cannot do. The trustworthiness of AI products should also be taken into consideration. Any future integration of AI should be accompanied by a lesson on myths and misconceptions regarding AI. Overall, most students perceive AI positively, but it is important to address misunderstandings and educate students about the realistic capabilities and limitations of AI. To summarise the practical implications from the themes, in order to enhance human-AI collaboration, students and teachers need to develop certain level of AI literacy.

5.5 Limitations

There were some limitations to this study. Firstly, the study was conducted within a specific geographic region and cultural context, which may limit the generalizability of the findings to other regions and cultures. The results were also derived from a single age group—15–16 year old students—and therefore do not represent all student groups engaged in complex problem-solving. Additionally, our use of the term 'AI' to describe the technology involved in this study may have activated certain preconceptions and biases among participants. This could have influenced their perceptions and interactions with the technology, potentially acting as a confounding factor in our analysis. To address these limitations, future research should incorporate multi-site studies involving diverse geographic regions and cultural contexts. This would provide a more comprehensive understanding of the dynamics of Human-AI teaming and its implications across different educational systems and cultural backgrounds. Furthermore, studies should encompass a broader range of age groups, including elementary and secondary school students, to understand the developmental nuances and educational stages affecting the efficacy of Human-AI teaming. In addition, future studies should consider the impact of terminology and preconceptions about AI on user interactions and perceptions. This could involve varying the descriptors used for the technology (e.g., tool, model) to assess if and how these labels influence user experience and perceptions.

6 Conclusion

Our study provides insights into students' perceptions and attitudes towards AI in education. The scientific contribution of this research lies in identifying the gap between students' expectations and the realistic capabilities of AI, thereby highlighting the necessity for targeted educational interventions. The findings suggest that students have unrealistic expectations of AI's capabilities, which highlights the need for educational interventions to improve their understanding of AI's potential and limitations. Specifically, this study contributes to the literature by emphasizing the importance of integrating Explainable AI (XAI) into educational contexts to enhance students' AI literacy. The practical implications of these findings indicate that co-creation of AI systems should involve conversations and explainability among teachers, students, and developers. For instance, Future research should include collaborative workshops where students actively participate in the design and evaluation of AI tools, fostering a deeper understanding and trust in AI technologies. The notion of Explainable AI, emerges as a pivotal concept in addressing these challenges. This offers a paradigm through which AI's decision-making processes can be rendered transparent and intelligible, thereby demystifying the technology and aligning it more closely with the users' cognitive and developmental stages. This approach is particularly pertinent in the context of our articipants, whose responses occasionally betrayed a lack of awareness of AI's current applications in everyday technologies. Overall, this study highlights the importance of addressing misunderstandings and educating students about the realistic capabilities and limitations of AI to ensure successful human-AI teaming. By incorporating these specific strategies and focusing on XAI, educators and developers can better align AI technologies with educational objectives, ultimately fostering more effective and meaningful human-AI collaborations.

7 Ethics

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The study was approved by the University of South Australia (No: 204417). The data that support the findings of this study are not publically available due to the confidential nature. The prcosses data are, however, available from the authors upon reasonable request and with the permission of The University of South Australia.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions. The authors declare no funding was received for this project.

Declarations

Conflicts of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Consent to Participate No consent was taken from participants as the interviews were conducted by the classroom teacher to inform classroom practices.

Consent to Publish Consent to publish was obtained from participants via the school.

References

Ahmad, M. I., Khordi-Moodi, M., & Lohan, K. S. (2020). Social robot for STEM education. In *Companion of the 2020 ACM/IEEE international conference on human–robot interaction* (pp. 90–92).

Ahmad, S. F., Alam, M. M., Rahmat, M. K., Mubarik, M. S., & Hyder, S. I. (2022). Academic and administrative role of artificial intelligence in education. *Sustainability, 14*(3), 1101.

Alam, A. (2021). Possibilities and apprehensions in the landscape of artificial intelligence in education. In *2021 International conference on computational intelligence and computing applications (ICCICA)* (pp. 1–8). IEEE.

AlShaikh, F., & Hewahi, N. (2021). Ai and machine learning techniques in the development of Intelligent Tutoring System: A review. In *2021 International conference on innovation and Intelligence for informatics, computing, and technologies (3ICT)* (pp. 403–410). IEEE.

Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI conference on human factors in computing systems*, 1–13. https://doi.org/10.1145/3290605.3300233

BaïDoo-Anu, D., & Owusu Ansah, L. (2023). Education in the Era of Generative Artificial Intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *Journal of AI*, *7*(1), 52–62. https://doi.org/10.61969/jai.1337500

Bayrak, A. E., McComb, C., Cagan, J., & Kotovsky, K. (2021). A strategic decision-making architecture toward hybrid teams for dynamic competitive problems. *Decision Support Systems, 144*, 113490. https://doi.org/10.1016/j.dss.2020.113490

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Braun, V., & Clarke, V. (2019). Reflecting on reflexive thematic analysis. *Qualitative Research in Sport, Exercise and Health, 11*(4), 589–597. https://doi.org/10.1080/2159676X.2019.1628806

Chen, X., Zou, D., Xie, H., Cheng, G., & Liu, C. (2022). Two decades of artificial intelligence in education. *Educational Technology & Society, 25*(1), 28–47.

Chiu, T. K. (2021). A holistic approach to the design of artificial intelligence (AI) education for K-12 schools. *TechTrends, 65*(5), 796–807. https://doi.org/10.1007/s11528-021-00637-1

Chiu, T. K., Meng, H., Chai, C.-S., King, I., Wong, S., & Yam, Y. (2021). Creation and evaluation of a pretertiary artificial intelligence (AI) curriculum. *IEEE Transactions on Education, 65*(1), 30–39.

Chounta, I.-A., Bardone, E., Raudsep, A., & Pedaste, M. (2021). Exploring teachers' perceptions of Artificial Intelligence as a tool to support their practice in Estonian K-12 education. *International Journal of Artificial Intelligence in Education*, 1–31.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement, 20*(1), 37–46. https://doi.org/10.1177/001316446002000104

Cooke, N. J., & Hilton, M. L. (2015). *Enhancing the effectiveness of team science*. ERIC.

Crompton, H., Jones, M. V., & Burke, D. (2022). Affordances and challenges of artificial intelligence in K-12 education: A systematic review. *Journal of Research on Technology in Education*, 1–21. https://doi.org/10.1080/15391523.2022.2121344

Dai, Y., Chai, C.-S., Lin, P.-Y., Jong, M.S.-Y., Guo, Y., & Qin, J. (2020). Promoting students' well-being by developing their readiness for the artificial intelligence age. *Sustainability, 12*(16), 6597.

de Laat, M., Joksimovic, S., & Ifenthaler, D. (2020). Artificial intelligence, real-time feedback and workplace learning analytics to support in situ complex problem-solving: A commentary. *The*

International Journal of Information and Learning Technology, 37(5), 267–277. https://doi.org/10.1108/IJILT-03-2020-0026

De Visser, E. J., Peeters, M. M., Jung, M. F., Kohn, S., Shaw, T. H., Pak, R., & Neerincx, M. A. (2020). Towards a theory of longitudinal trust calibration in human–robot teams. *International Journal of Social Robotics, 12*(2), 459–478.

Demir, K., & Güraksin, G. E. (2022). Determining middle school students' perceptions of the concept of artificial intelligence: A metaphor analysis. *Participatory Educational Research, 9*(2), 297–312.

Dilmurod, R., & Fazliddin, A. (2021). Prospects for the introduction of artificial intelligence technologies in higher education. *ACADEMICIA: An International Multidisciplinary Research Journal*, *11*(2), 929–934.

Druga, S., Williams, R., Breazeal, C., & Resnick, M. (2017). 'Hey Google is it ok if I eat you?' Initial explorations in child-agent interaction. 595–600.

Druga, S., Williams, R., Park, H. W., & Breazeal, C. (2018). How smart are the smart toys? Children and parents' agent interaction and intelligence attribution. 231–240.

Ekbia, H. R., & Nardi, B. A. (2017). *Heteromation, and other stories of computing and capitalism*. The MIT Press.

Ezer, N., Bruni, S., Cai, Y., Hepenstal, S. J., Miller, C. A., & Schmorrow, D. D. (2019). Trust engineering for human-AI teams. In *Proceedings of the human factors and ergonomics society annual meeting* (vol. 63, no. 1, pp. 322–326). Sage CA: Los Angeles, CA: SAGE Publications.

Fast, E., & Horvitz, E. (2017). Long-term trends in the public perception of artificial intelligence. In *Proceedings of the AAAI conference on artificial intelligence* (vol. 31, no. 1).

Ferguson, R., Brasher, A., Clow, D., Cooper, A., Hillaire, G., Mittelmeier, J., Rienties, B., Ullmann, T., & Vuorikari, R. (2016). *Research evidence on the use of learning analytics: Implications for education policy*.

Fiore, S. M. (2021). Interdisciplinary models and frameworks for the study of artificial social intelligence. *65*(1), 658–659.

Freeman, A., Becker, S. A., & Cummins, M. (2017). *NMC/CoSN horizon report: 2017 K*. The New Media Consortium.

Gunning, D., & Aha, D. (2019). DARPA's explainable artificial intelligence (XAI) program. *AI Magazine, 40*(2), 44–58.

Holmes, W., Bialik, M., & Fadel, C. (2019). *Artificial intelligence in education promises and implications for teaching and learning*. Center for Curriculum Redesign.

Holstein, K., & Aleven, V. (2022). Designing for human–AI complementarity in K-12 education. *AI Magazine, 43*(2), 239–248. https://doi.org/10.1002/aaai.12058

Holzinger, A., Saranti, A., Molnar, C., Biecek, P., & Samek, W. (2022). *Explainable AI methods-a brief overview*. 13–38.

Hsu, H. L., Chen, H. H. J., & Todd, A. G. (2021). Investigating the impact of the Amazon Alexa on the development of L2 listening and speaking skills. *Interactive Learning Environments,* 1–14.

Jeong, H., Hmelo-Silver, C. E., & Jo, K. (2019). Ten years of computer-supported collaborative learning: A meta-analysis of CSCL in STEM education during 2005–2014. *Educational Research Review, 28*, 100284.

Joksimovic, S., Ifenthaler, D., Marrone, R., De Laat, M., & Siemens, G. (2023). Opportunities of artificial intelligence for supporting complex problem-solving: Findings from a scoping review. *Computers and Education: Artificial Intelligence*, 100138.

Kaput, M. (2022). *What skills remain uniquely human in an AI-First World? [Video]*. What Skills Remain Uniquely Human in an AI-First World? [Video]. https://www.marketingaiinstitute.com/blog/what-skills-remain-uniquely-human-in-an-ai-first-world

Karampelas, A. (2020). Artificial Intelligence and Machine Learning in the STEAM classroom: Analysis of performance data and reflections of international high school students. *Hellenic Journal of STEM Education, 1*(2), 59–66.

Keles, P. U., & Aydin, S. (2021). University students' perceptions about artificial intelligence. *Shanlax International Journal of Education, 9*, 212–220.

Lodge, J. M., Yang, S., Furze, L., & Dawson, P. (2023). It's not like a calculator, so what is the relationship between learners and generative artificial intelligence? *Learning: Research and Practice*, *9*(2), 117–124.

Luckin, R., & Holmes, W. (2016). *Intelligence unleashed: An argument for AI in education*.

Lyons, J. B., Wynne, K. T., Mahoney, S., & Roebke, M. A. (2019). Trust and human-machine teaming: A qualitative study. In *Artificial intelligence for the internet of everything* (pp. 101–116). Elsevier.

Markauskaite, L., Marrone, R., Poquet, O., Knight, S., Martinez-Maldonado, R., Howard, S., Tondeur, J., De Laat, M., Buckingham Shum, S., Gašević, D., & Siemens, G. (2022). Rethinking the entwinement between artificial intelligence and human learning: What capabilities do learners need for a world with AI? *Computers and Education: Artificial Intelligence, 3*, 100056. https://doi.org/10.1016/j.caeai.2022.100056

Marrone, R., Taddeo, V., & Hill, G. (2022). Creativity and artificial intelligence: A student perspective. *Journal of Intelligence, 10*(3), 65. https://doi.org/10.3390/jintelligence10030065

McCall, F., Hussein, A., Petraki, E., Elsawah, S., & Abbass, H. (2021). *Towards a systematic educational framework for human–machine teaming*. 375–382.

McNeese, N. J., Demir, M., Chiou, E. K., & Cooke, N. J. (2021). Trust and team performance in human–autonomy teaming. *International Journal of Electronic Commerce, 25*(1), 51–72.

Minn, S. (2022). AI-assisted knowledge assessment techniques for adaptive learning environments. *Computers and Education: Artificial Intelligence*, 100050.

Mousavinasab, E., Zarifsanaiey, N., R. Niakan Kalhori, S., Rakhshan, M., Keikha, L., & Ghazi Saeedi, M. (2021). Intelligent tutoring systems: A systematic review of characteristics, applications, and evaluation methods. *Interactive Learning Environments*, *29*(1), 142–163.

Musick, G., O'Neill, T. A., Schelble, B. G., McNeese, N. J., & Henke, J. B. (2021). What happens when humans believe their teammate is an AI? An investigation into humans teaming with autonomy. *Computers in Human Behavior, 122*, 106852.

Papadopoulos, I., Lazzarino, R., Miah, S., Weaver, T., Thomas, B., & Koulouglioti, C. (2020). A systematic review of the literature regarding socially assistive robots in pre-tertiary education. *Computers & Education, 155*, 103924.

Pearl, J. (2019). The limitations of opaque learning machines. *Possible Minds, 25*, 13–19.

Renz, A., & Vladova, G. (2021). Reinvigorating the discourse on human-centered artificial intelligence in educational technologies. *Technology Innovation Management Review*, *11*(5).

Salas, E., Cooke, N. J., & Rosen, M. A. (2008). On teams, teamwork, and team performance: Discoveries and developments. *Human Factors: THe Journal of the Human Factors and Ergonomics Society, 50*(3), 540–547. https://doi.org/10.1518/001872008X288457

Schaefer, K. E., Perelman, B., Rexwinkle, J., Canady, J., Neubauer, C., Waytowich, N., Larkin, G., Cox, K., Geuss, M., & Gremillion, G. (2021). Human-autonomy teaming for the tactical edge: The importance of humans in artificial intelligence research and development. In *Systems engineering and artificial intelligence* (pp. 115–148). Springer.

Seeber, I., Bittner, E., Briggs, R. O., De Vreede, G.-J., De Vreede, T., Druckenmiller, D., Maier, R., Merz, A. B., Oeste-Reiß, S., & Randrup, N. (2018). *Machines as teammates: A collaboration research agenda*.

Seo, K., Tang, J., Roll, I., Fels, S., & Yoon, D. (2021). The impact of artificial intelligence on learner–instructor interaction in online learning. *International Journal of Educational Technology in Higher Education, 18*, 1–23.

Shaffer, D. W., Collier, W., & Ruis, A. R. (2016). A tutorial on epistemic network analysis: Analyzing the structure of connections in cognitive, social, and interaction data. *Journal of Learning Analytics*, *3*(3), 9–45. https://doi.org/10.18608/jla.2016.33.3

Shaffer, D. W., Hatfield, D., Svarovsky, G. N., Nash, P., Nulty, A., Bagley, E., Frank, K., Rupp, A. A., & Mislevy, R. (2009). Epistemic network analysis: A prototype for 21st-century assessment of learning. *International Journal of Learning and Media, 1*(2), 33–53. https://doi.org/10.1162/ijlm.2009.0013

Siemens, G., Marmolejo-Ramos, F., Gabriel, F., Medeiros, K., Marrone, R., Joksimovic, S., & de Laat, M. (2022). Human and artificial cognition. *Computers and Education: Artificial Intelligence, 3*, 100107.

Staker, H., & Horn, M. B. (2014). Classifying K–12 blended learning. *Blended Learning Research Perspectives*, 287–303.

Textor, C., Zhang, R., Lopez, J., Schelble, B. G., McNeese, N. J., Freeman, G., Pak, R., Tossell, C., & de Visser, E. J. (2022). Exploring the relationship between ethics and trust in human-artificial intelligence teaming: A mixed methods approach. *Journal of Cognitive Engineering and Decision Making, 16*(4), 252–281.

Touretzky, D., Gardner-McCune, C., Martin, F., & Seehorn, D. (2019). *Envisioning AI for K-12: What should every child know about AI? 33*(01), 9795–9799.

Van Brummelen, J., Tabunshchyk, V., & Heng, T. (2021). "Alexa, Can I Program You?": Student perceptions of conversational artificial intelligence before and after programming alexa. *Interaction Design and Children*, 305–313. https://doi.org/10.1145/3459990.3460730

Vogt, P., van den Berghe, R., De Haas, M., Hoffman, L., Kanero, J., Mamus, E., Montanier, J.-M., Oranç, C., Oudgenoeg-Paz, O., & García, D. H. (2019). *Second language tutoring using social robots: A large-scale study*. 497–505.

Walter, Y. (2024). Embracing the future of Artificial Intelligence in the classroom: The relevance of AI literacy, prompt engineering, and critical thinking in modern education. *International Journal of Educational Technology in Higher Education, 21*(1), 15.

Wang, X., Pang, H., Wallace, M. P., Wang, Q., & Chen, W. (2022). Learners' perceived AI presences in AI-supported language learning: A study of AI as a humanized agent from community of inquiry. *Computer Assisted Language Learning, 37*(4), 814–840. https://doi.org/10.1080/09588221.2022.2056203

Williams, J., Fiore, S. M., & Jentsch, F. (2022). Supporting artificial social intelligence with theory of mind. *Frontiers in Artificial Intelligence*, *5*.

Wilson, H. J., & Daugherty, P. R. (2018). Collaborative intelligence: Humans and AI are joining forces. *Harvard Business Review, 96*(4), 114–123.

Yin, M., Wortman Vaughan, J., & Wallach, H. (2019). *Understanding the effect of accuracy on trust in machine learning models*. 1–12.

Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education – where are the educators?. *Abstract International Journal of Educational Technology in Higher Education, 16*(1). https://doi.org/10.1186/s41239-019-0171-0

Zhang, R., McNeese, N. J., Freeman, G., & Musick, G. (2021). 'An ideal human' expectations of AI teammates in human-AI teaming. *Proceedings of the ACM on Human-Computer Interaction, 4*(CSCW3), 1–25.